

DAY 1 EXCERISE

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Course Name: CSA1614 Data Warehousing and Data Mining

Q1.

```
1 age <- c(5, 15, 20, 50, 80, 110)
2 frequency <- c(200, 450, 300, 1500, 700, 44)
3 median(age)
4 median(frequency)
```

Console **Terminal** × **Background Jobs** ×

R 4.6.1 · ~/

R version 4.6.1 (2026-06-24 ucrt) -- "Happy Hop"
Platform: x86_64-w64-mingw32/x64

```
> median(age)
[1] 35
> median(frequency)
[1] 375
```

Q2.

```
1 age <- c(13,15,16,16,19,20,20,21,22,22,25,25,25,25,30,33,33,35,35,35,35,36,40,45,45)
2 mean(age)
3 median(age)
4 mode_age <- names(table(age))[table(age)==max(table(age))]
5 mode_age
6 range(age)
7 quantile(age,.25)
8 quantile(age,.75)
```

Console Terminal Background Jobs

R version 4.6.1 (2026-06-24 ucrt) -- "Happy Hop"
Platform: x86_64-w64-mingw32/x64

```
> mean(age)
[1] 29.96296
> median(age)
[1] 25
> mode_age
[1] "25" "35"
> range(age)
[1] 13 70
> quantile(age,.25)
25%
20.5
> quantile(age,.75)
75%
35
```

Q3.

```
1 x <- c(200, 300, 400, 600, 1000)
2 min_max_norm <- (x-min(x))/(max(x)-min(x))
3 min_max_norm
4 z_score_norm <- (x-mean(x))/sd(x)
5 z_score_norm
```

Console Terminal Background Jobs

R version 4.6.1 (2026-06-24 ucrt) -- "Happy Hop"
Platform: x86_64-w64-mingw32/x64

```
> min_max_norm
[1] 0.000 0.125 0.250 0.500 1.000
> z_score_norm
[1] -0.9486833 -0.6324555 -0.3162278 0.3162278 1.5811388
```

Q4.

```
1 data <- c(11,13,13,15,15,16,19,20,20,20,21,21,22,23,24,30,40,45,45,45,71,72,75)
2 bins <- 5
3 bin_indices <- cut(data, bins)
4 tapply(data, bin_indices, mean)
5 tapply(data, bin_indices, median)
6 tapply(data, bin_indices, function(x) c(min(x), max(x)))
```

Console Terminal x Background Jobs x

R 4.6.1 · ~/

R version 4.6.1 (2026-06-24 ucrt) -- "Happy Hop"
Platform: x86_64-w64-mingw32/x64

```
> tapply(data, bin_indices, mean)
(11,23.8] (23.8,36.6] (36.6,49.4] (49.4,62.2] (62.2,75]
18.30769 27.00000 43.75000 NaN 72.75000
> tapply(data, bin_indices, median)
20 27 45 NaN 72.5
> tapply(data, bin_indices, function(x) c(min(x), max(x)))
(11,23.8] 13 23
(23.8,36.6] 24 30
(36.6,49.4] 40 45
(49.4,62.2] numeric(0)
(62.2,75] 71 75
```

Q5.

RStudio

FileEditCodeViewPlotsSessionBuildDebugProfileToolsHelp

Untitled5*Source on SaveRunSource

```
1 age <- c(23,23,27,27,39,41,47,49,50,52,
2         54,54,56,57,58,58,60,61)
3 fat <- c(9.5,26.5,7.8,17.8,31.4,25.9,27.4,
4         27.2,31.2,34.6,42.5,28.8,33.4,30.2,
5         34.1,32.9,41.2,35.7)
6 mean(age)
7 median(age)
8 sd(age)
9 mean(fat)
10 median(fat)
11 sd(fat)
12 # boxplot
13 boxplot(age,fat)
14 # scatter plot
15 scatter.smooth(age,fat)
```

EnvironmentHistoryConnectionsTutorial

Global Environment

Environment is empty

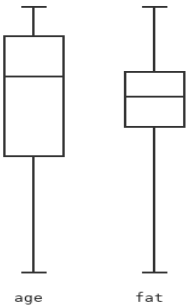
ConsoleTerminalBackground Jobs

R 4.6.1 ~/

```
> mean(age)
[1] 46.44444
> median(age)
[1] 51
> sd(age)
[1] 13.21862
> mean(fat)
[1] 28.78333
> median(fat)
[1] 30.7
> sd(fat)
[1] 9.254395
>
```

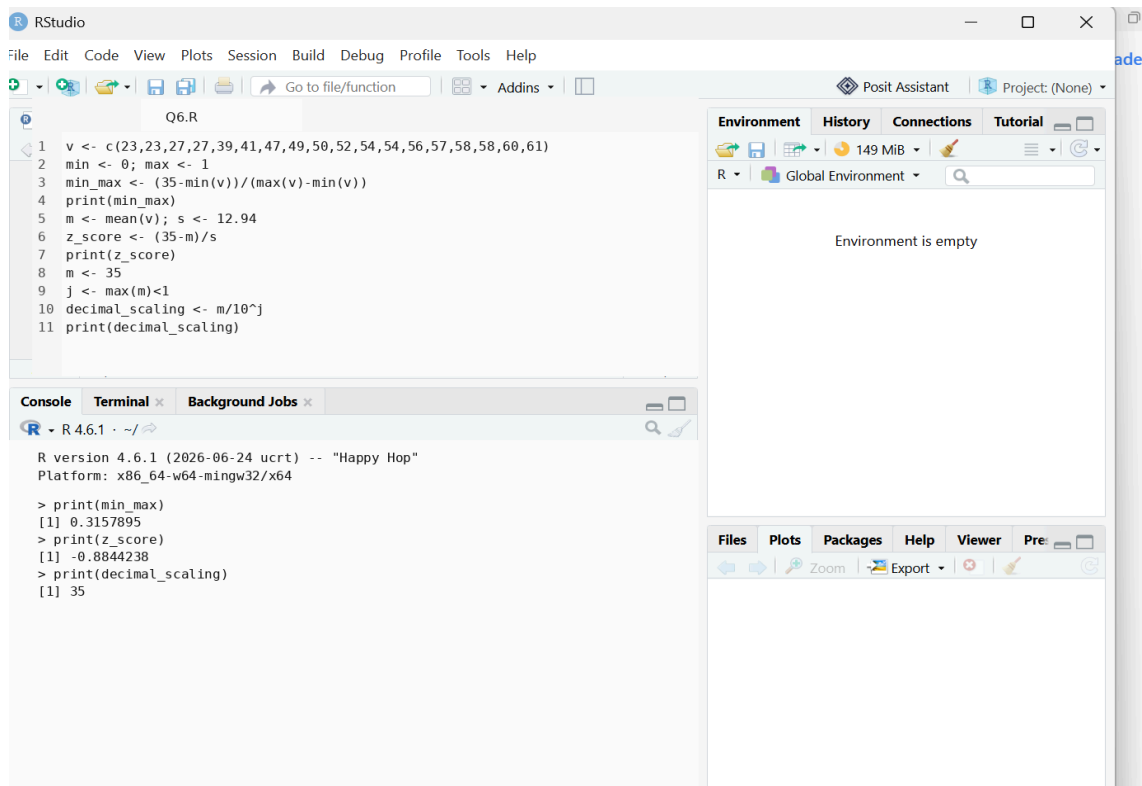
FilesPlotsPackagesHelpViewer

Boxplot (age, fat)



Variable	Min	Q1	Median	Q3	Max
age	23	46	51	58	61
fat	7.8	26.5	30.7	34.6	42.5

Q6.



The RStudio interface for Q6 shows a script file named 'Q6.R' with the following code:

```
1 v <- c(23,23,27,27,39,41,47,49,50,52,54,54,56,57,58,58,60,61)
2 min <- 0; max <- 1
3 min_max <- (35-min(v))/(max(v)-min(v))
4 print(min_max)
5 m <- mean(v); s <- 12.94
6 z_score <- (35-m)/s
7 print(z_score)
8 m <- 35
9 j <- max(m)<1
10 decimal_scaling <- m/10^j
11 print(decimal_scaling)
```

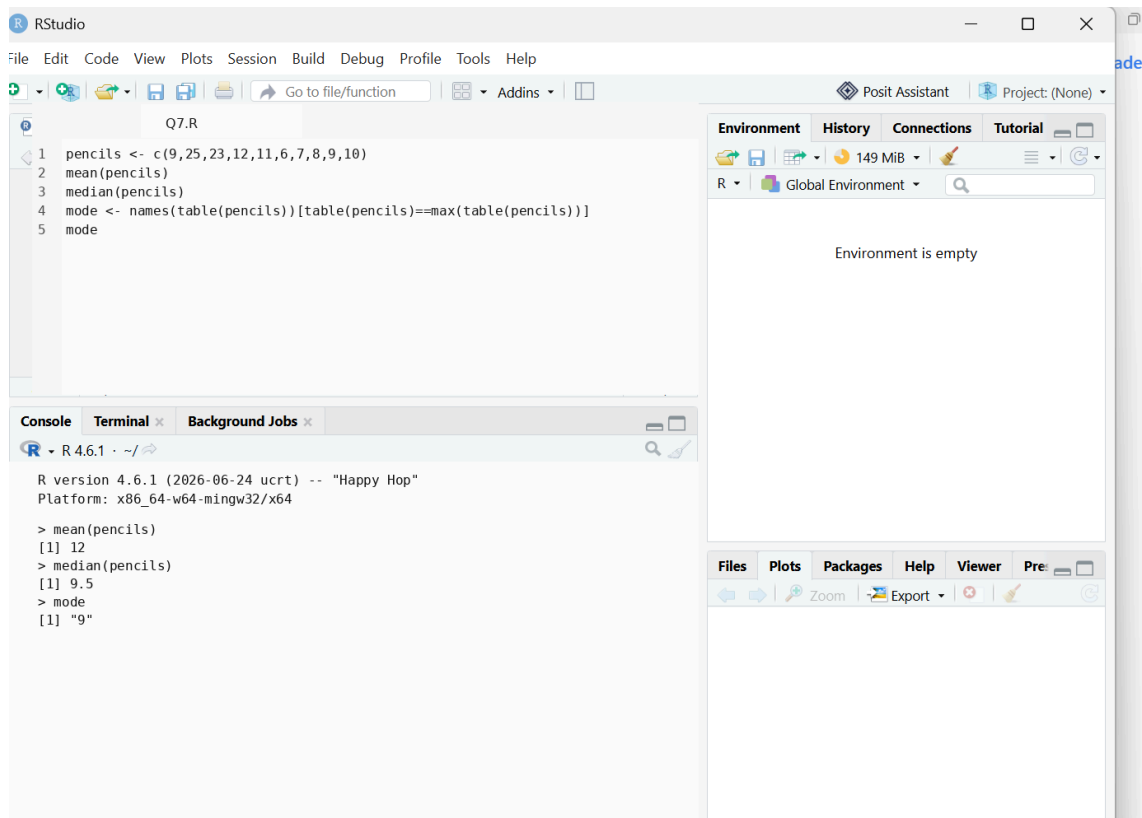
The console output shows the results of the code execution:

```
R version 4.6.1 (2026-06-24 ucrt) -- "Happy Hop"
Platform: x86_64-w64-mingw32/x64

> print(min_max)
[1] 0.3157895
> print(z_score)
[1] -0.8844238
> print(decimal_scaling)
[1] 35
```

The Environment pane on the right shows 'Global Environment' and 'Environment is empty'.

Q7



The RStudio interface for Q7 shows a script file named 'Q7.R' with the following code:

```
1 pencils <- c(9,25,23,12,11,6,7,8,9,10)
2 mean(pencils)
3 median(pencils)
4 mode <- names(table(pencils))[table(pencils)==max(table(pencils))]
5 mode
```

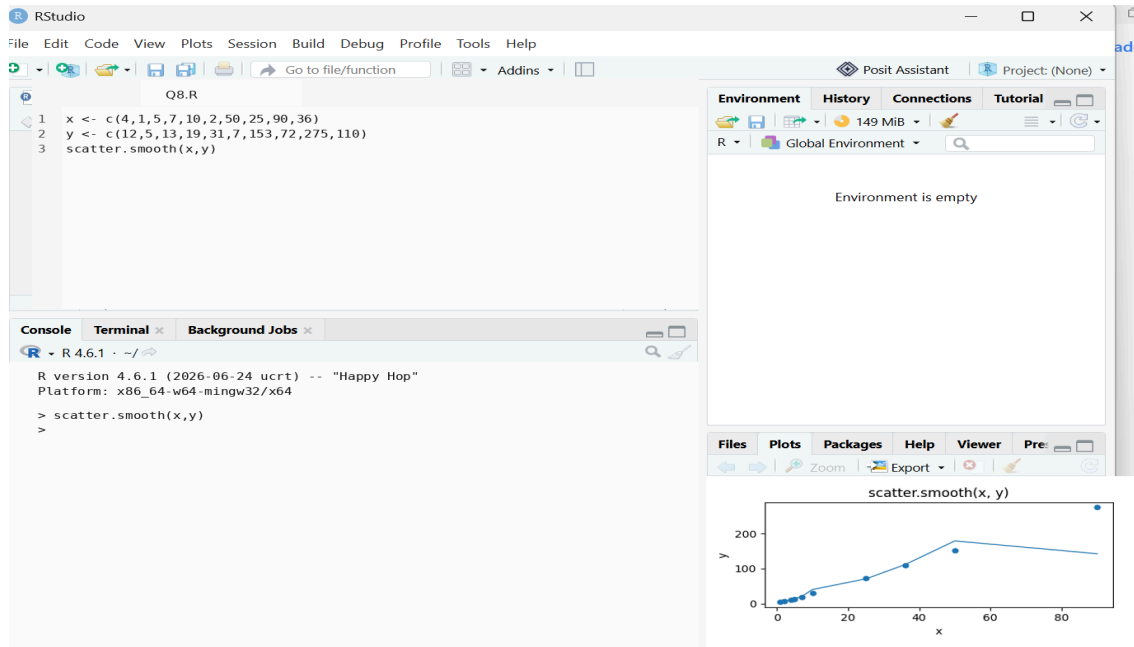
The console output shows the results of the code execution:

```
R version 4.6.1 (2026-06-24 ucrt) -- "Happy Hop"
Platform: x86_64-w64-mingw32/x64

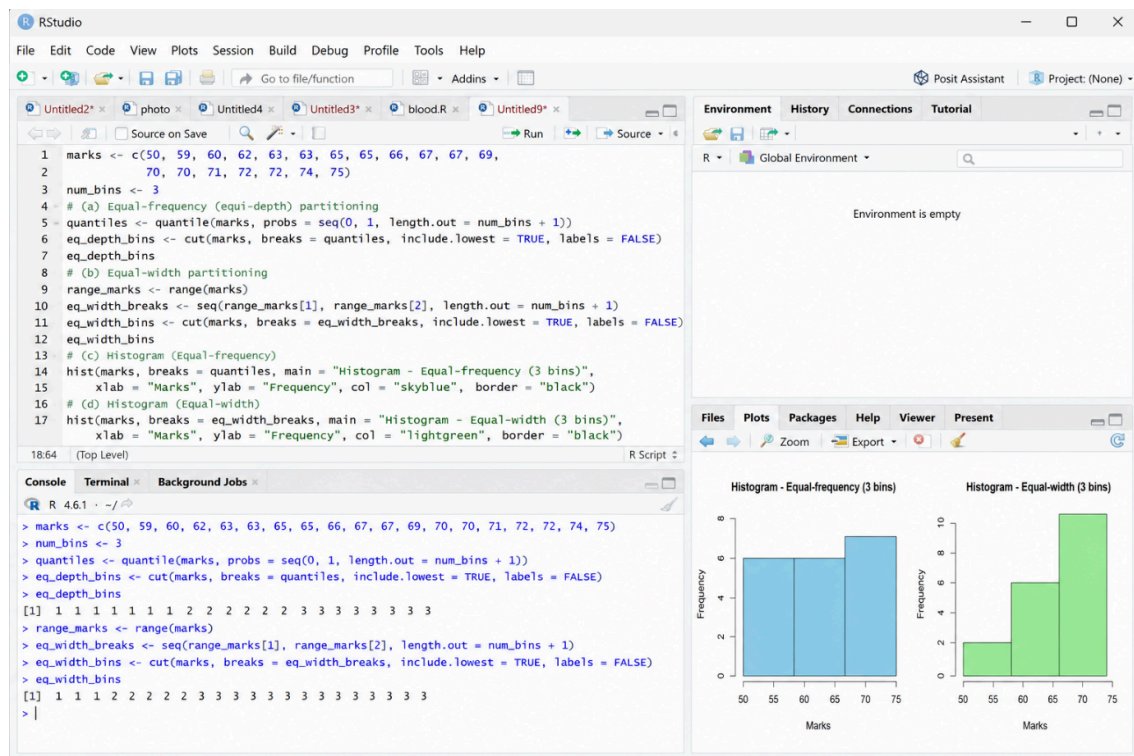
> mean(pencils)
[1] 12
> median(pencils)
[1] 9.5
> mode
[1] "9"
```

The Environment pane on the right shows 'Global Environment' and 'Environment is empty'.

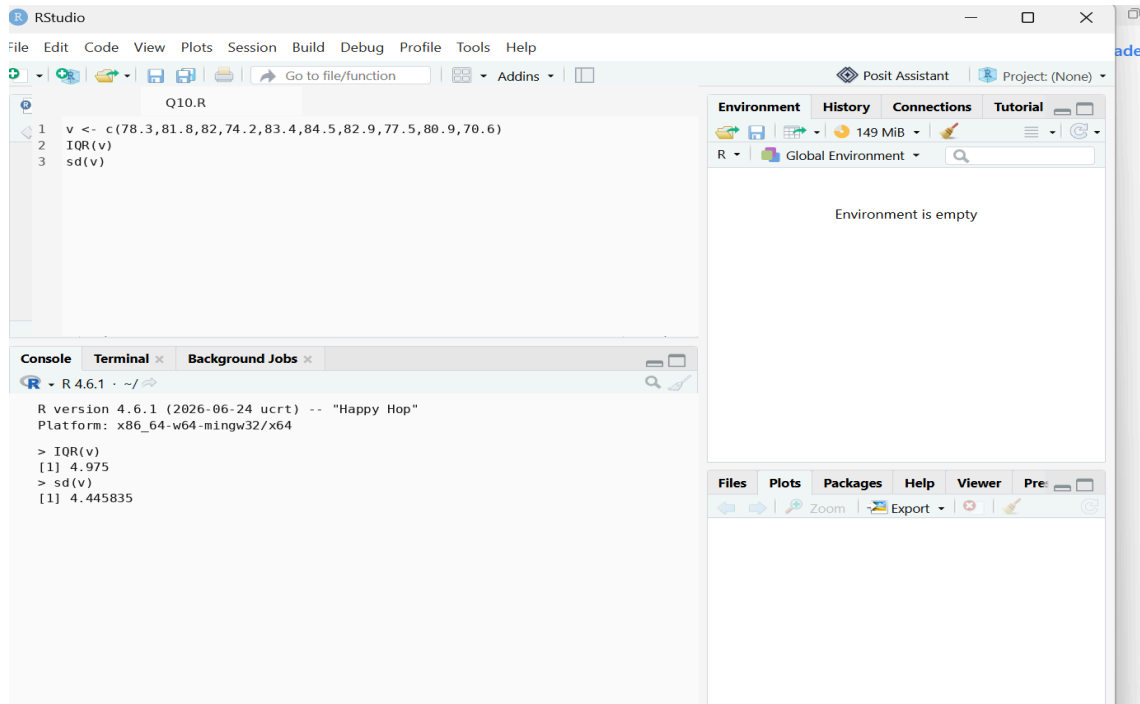
Q8



Q9



Q10



Q11

